

Variational Autoencoder based Latent Factor Decoding of Multichannel EEG for Emotion Recognition

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Abstract—Robust cross-subject emotion recognition based on multichannel EEG has always been a hard work. In this work, we hypothesize there exists default brain variables across subjects in emotional processes. Hence, the states of the latent variables that related to emotional processing must contribute to building robust recognition models. We propose to utilize variational autoencoder (VAE) to determine the latent factors from the multichannel EEG. Through sequence modeling method, we examine the emotion recognition performance based on the learnt latent factors. The performance of the proposed methodology is verified on two public datasets (DEAP and SEED), and compared with traditional matrix factorization based (ICA) and autoencoder based (AE) approaches. Experimental results demonstrate that neural network is suitable for unsupervised EEG modeling and our proposed emotion recognition framework achieves the state-of-the-art performance. As far as we know, it is the first work that introduces VAE into multichannel EEG decoding for emotion recognition.

Index Terms—Affective Computing, Latent Factor Decoding, Emotion Recognition, EEG, Variational Autoencoder.

I. INTRODUCTION

Affective Computing starts becoming an active research topic. Its main objective is exploring effective approaches in recognizing person's emotions automatically by utilizing explicit or implicit body information, particularly the multichannel EEG (electroencephalograph). Nevertheless, there exists some major problems in multichannel EEG based emotion recognition that needs to deal with, such as the poor

generalization of data across subjects and the limitations in designing and extracting handcrafted emotion-related EEG features. Further, in medical data mining tasks, it is still a problem to acquire enough manually labeled data for training supervised models. How to fully utilize the limited data to enhance the model performance is worth of exploration. Hence, unsupervised and handcrafted feature non-dependent methods are worth deep exploration.

In this work, we utilize the findings of prior related works [7] and raise the hypothesis that though differences exist between individuals, there exists intrinsic default variables (e.g., brain networks or intracranial current sources) that take part in emotional processes. The status information of the latent factors contains emotion-related information that contribute to estimating the emotional status. Hence, how to effectively and precisely infer the latent factors from the external multichannel EEGs is the core issue that we concern in this work.

II. METHODS

A. VAE based Decoding Approach

Latent factor decoding from brain activity signal is a key tool to study the cognitive task performance. The independent component analysis (ICA) is the most commonly used method. The utilization of ICA has been limited by its flexibility and representation ability [2]. Hence, its effectiveness in cross-subject EEG decoding is questionable. Very recently,

the VAE is introduced as a powerful DL model for some problem scenarios that needs modeling the data's probability distribution [3]. Compared with traditional AE model, the VAE model provides a closed-form representation of the distribution underlying the input data. The mechanism of the VAE model is illustrated in Figure 1.

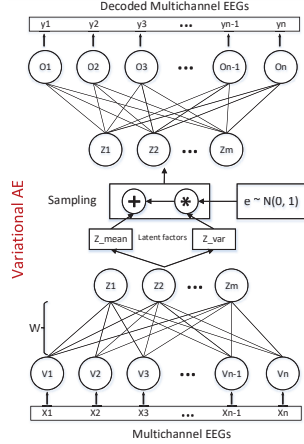


Fig. 1. The VAE based multichannel EEG fusion and latent factor decoding method.

It hypothesizes all the data are generated by one random process, which involves an unobservable latent variable z . The latent variable is generated from one prior distribution $p_\theta(z)$, and the x is determined by the conditional distribution $p_\theta(x|z)$. Both the parameters θ and the latent variable z are unknown to us. The direct inference of the latent variable $p_\theta(z|x)$ is intractable. Hence, in the design of the VAE, one recognition model $q_\phi(z|x)$ is introduced to approximate the true posterior $p_\theta(z|x)$. The VAE utilizes the probabilistic encoder structure to encode the input into latent variables ($q_\phi(z|x)$), and it further utilizes the probabilistic decoder structure to map the latent variables to reconstructed input ($p_\theta(x|z)$). The optimization objective of the VAE is expressed in Formula 1:

$$\max \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x) || p_\theta(z)) \quad (1)$$

, which is also referred to as the variational lower bound.

B. Contextual Learning from Latent Factors for Emotion Recognition

The recurrent neural network (RNN) has the ability of accumulating useful information at each time step. It is important when we do not know which moment plays the most important role in the subject's final evaluation of the specific emotion they experienced in a trial. In view of this,

we consider adopting the popular LSTM unit based RNN model (LSTM-RNN) to do sequence modeling on the decoded latent factor sequences, meanwhile the subjects' emotional status can be estimated. We feed the multichannel EEGs into the 'encoder' part of the trained models to obtain the corresponding latent factor sequences, namely the independent components (ICs). As the high sampling rate signal will lead to the high computational cost in sequence modeling. Here we need sampling from the entire ICs to construct samples for RNN training. The entire method framework is illustrated in the flow chart as in Figure 2.

III. EXPERIMENTS

A. Experimental Dataset

We examine the proposed approach on two publicly accessible datasets, including **DEAP**¹ and **SEED**². DEAP includes 32-channel EEG data collected from 32 subjects. SEED includes 62-channel EEG data collected from 15 subjects. The DEAP are divided into positive and negative samples according to the ratings on emotional dimensions of Valence and Arousal. A sample with score over 5 is considered a positive class, otherwise is considered a negative class. The SEED dataset has pre-defined negative and positive emotional classes for the samples that we do not need conduct label processing.

B. Results and Discussion

1) *Evaluation on EEG modeling and reconstruction:* As shown in Figure 3, we illustrate the performance of each method in reconstructing the multichannel EEGs through a channel layout heatmap format (the number of latent factors is set as half the number of electrode channels, DEAP: 16 latent factors, SEED: 31 latent factors). The mean Pearson correlation coefficient over all the subjects is presented. The greater the value, the hotter the color.

It can be seen that the frequently used ICA-based method is obviously inferior to other methods on the whole, and there exists significant performance imbalance in different brain regions. It indicates the ICA based EEG modeling approach is not robust compared to the neural network based approaches. Besides, the reconstruction performance under different assumed number of latent factors is also of interest. As shown in Figure 4, the reconstruction performance of the ICA-based method is always much lower than the neural

¹<http://www.eecs.qmul.ac.uk/mmv/datasets/deap/index.html>

²<http://bcmi.sjtu.edu.cn/seed/index.html>

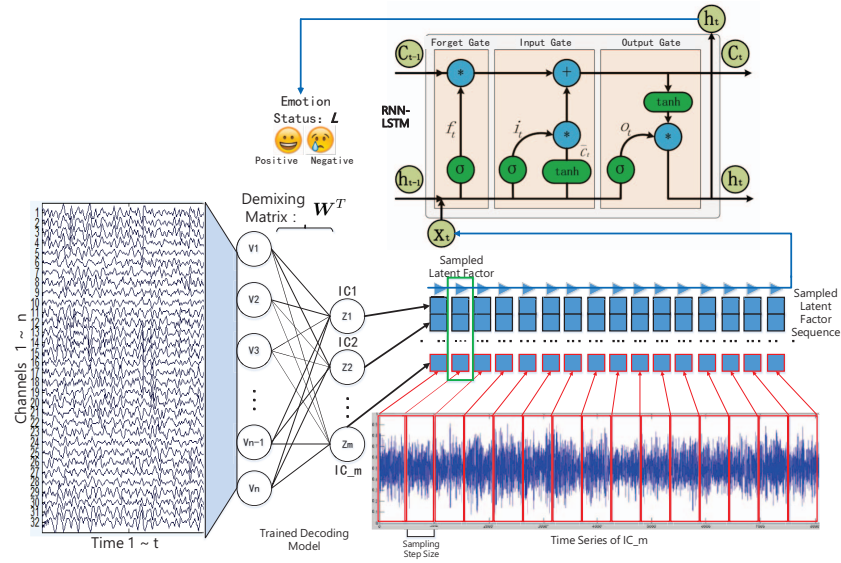


Fig. 2. The latent EEG factors and recurrent neural network based emotion recognition approach framework.

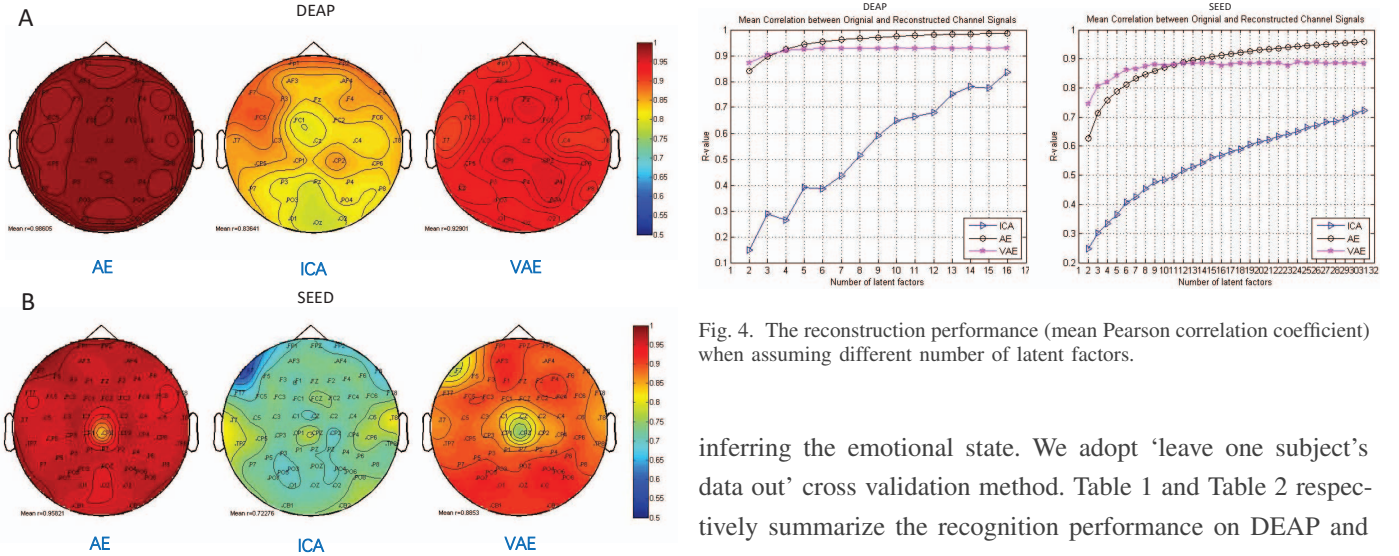


Fig. 3. Pearson correlation coefficient between the reconstructed EEG signal and the original EEG signal for each channel.

network based modeling method. Besides, the VAE based method can achieve better reconstruction performance than other methods with fewer hidden layer node settings, which indicates that the method has the ability to mine the most important factors from multichannel EEGs.

2) Evaluation on latent factor based emotion recognition:

It is necessary to apply pattern recognition methods on those mined factors, and conduct comparison based on the recognition performance. Specifically, the LSTM-RNN takes charge of modeling the latent factor sequence decoded, as well as

Fig. 4. The reconstruction performance (mean Pearson correlation coefficient) when assuming different number of latent factors.

inferring the emotional state. We adopt 'leave one subject's data out' cross validation method. Table 1 and Table 2 respectively summarize the recognition performance on DEAP and SEED. Recognition performance is measured and compared with state-of-the-art methods using F1 score.

It is found from the table that the ICA-based method on both datasets exceeds 0.5, indicating that even the traditional ICA based method can still decipher emotion related information from the multichannel EEG. It indicates that the latent factor decoding combined with sequence modeling based approach is suitable for emotion recognition from multichannel EEG. However, the ICA based approach does not perform as well as other neural network based approaches, confirming the conclusion that ICA has limitations in representation ability [2]. The VAE based method almost achieves the state-of-art performance on both datasets. It embodies the advantages of VAE in unsupervised modeling of brain neural signals.

TABLE I
PERFORMANCE COMPARISON BETWEEN THIS WORK AND BASELINE METHODS (DEAP)

Approach/Model	Performance	
	Valence	Arousal
CNN + channel-frequency two-dimensional spectral representation [9]	0.6459	0.6661
Ontology-based storage and representation, feature selection and decision tree based recognition method [1]	0.6783	0.6896
Integrated classifier based on multi-layer stacking autoencoder combined with time domain features and PSD features [8]	0.7243	0.6901
Generative adversarial network (WGANDA) based transfer learning combined with differential entropy feature [6]	0.6799	0.6685
ICA+RNN	0.6303	0.6418
AE+RNN	0.6783	0.6944
VAE+RNN	0.7167	0.7243

TABLE II
PERFORMANCE COMPARISON BETWEEN THIS WORK AND BASELINE METHODS (SEED)

Approach/Model	Performance	
	Valence	Arousal
CNN + channel-frequency two-dimensional spectral representation [9]	0.7604	0.8020
Extracting differential entropy features to construct 2D sparse graph representation, then combining CNN for classification [5]	0.8020	0.7247
Transfer learning methods combined with differential entropy features and logistic regression based classification [4]	0.7247	
ICA+RNN	0.6994	0.7452
AE+RNN	0.7452	0.8429
VAE+RNN	0.8429	

Although the AE has shown excellent ability in reconstructing the multichannel EEG, its decoded factors are less accurate reflection of the brain cognitive state.

IV. CONCLUSION

In this work, artificial neural network is used for unsupervised modeling of the state space of the latent factors from the multichannel EEGs, and RNN based supervised sequence modeling is further performed on the decoded latent factor sequences to mine the emotion related information, which is used for inferring the emotional states. We verify that the neural network models are more suitable for modeling and decoding brain neural signals than the independent component analysis (ICA) method, which is widely used in brain cognitive research. Specifically, The decoded latent factors from VAE contain the relevant and effective information for emotional state inference. This approach is promising in diagnosing depression, Alzheimer's disease, etc., whose specific brain functional networks may have been altered or aberrant.

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REFERENCES

- [1] Jing Chen, Bin Hu, and Philip Moore. Electroencephalogram-based emotion assessment system using ontology and data mining techniques. *Applied Soft Computing*, 30:663–674, 2015.
- [2] R. A Choudrey. Variational methods for bayesian independent component analysis. *University of Oxford*, 2002.
- [3] D Kingma and M Welling. Auto-encoding variational bayes. *international conference on learning representations*, 2014.
- [4] Zirui Lan, Olga Sourina, and Lipo Wang. Domain adaptation techniques for eeg-based emotion recognition: a comparative study on two public datasets. *IEEE Transactions on Cognitive and Developmental Systems*, 11(1):85–94, 2018.
- [5] Jinpeng Li, Zhaoxiang Zhang, and Huiguang He. Hierarchical convolutional neural networks for eeg-based emotion recognition. *Cognitive Computation*, 10(2):368–380, 2018.
- [6] Yun Luo, Si-Yang Zhang, and Wei-Long Zheng. Wgan domain adaptation for eeg-based emotion recognition. In *International Conference on Neural Information Processing*, pages 275–286. Springer, 2018.
- [7] Katherine Vytal and Stephan Hamann. Neuroimaging support for discrete neural correlates of basic emotions: A voxel-based meta-analysis. *Journal of Cognitive Neuroscience*, 22(12):2864–2885, 2014.
- [8] Z. Yin, M. Zhao, and Y. Wang. Recognition of emotions using multimodal physiological signals and an ensemble deep learning model. *Computer Methods and Programs in Biomedicine*, 140(C):93–110, 2017.
- [9] Guangliang Yu, Xiang Li, and Dawei Song. Encoding physiological signals as images for affective state recognition using convolutional neural networks. In *2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 812–815. IEEE, 2016.